Neural-Symbolic NLP: Bridging Theory and Practice

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Background

Symbolic models are good at explicit reasoning but break down easily once real-world data are taken into consideration, while neural networks show robust performance but lack in interpretability and systematic reasoning. The advent of large language models has increased interest in combining these two approaches.

The primary goal of this Research Topic is to explore the intersection of neural and symbolic approaches in NLP. It aims to bring together theoretical insights, practical implementations, and empirical evaluations, in order to create a comprehensive exploration of whether neural-symbolic NLP can advance our understanding of language, as well as our ability to create more reliable, efficient, and interpretable NLP systems.

Three core themes are proposed:

1. Neural-Symbolic Integration

The theme explores fundamental issues in combining neural and symbolic approaches:

- Architectures and methodologies for NeSy integration: Techniques to merge neural and symbolic systems.
- LLM-symbolic adaptation: Investigating

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performance and effectiveness of neural-symbolic systems in NLP contexts.

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2. Linguistic Theory and Neural NLP

The theme investigates connections between linguistic theory and neural NLP:

- Linguistic hypothesis testing: neural/NeSy models for testing linguistic theories.
- Language acquisition: neural/NeSy models for language acquisition
- Formal linguistic theories for neural systems: integration of linguistic theories into neural network architectures.
- Compatibility: the compatibility of linguistic theories and neural computational models.

3. Transparency and Explainability

The theme will cover critical challenges that can be potentially improved when using NeSy systems:

- Verification: reliability of NLP systems.
- Efficiency: Ways to measure the computational and resource efficiency of NeSy models
- Interpretability: why and how NeSy systems are more interpretable

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Keywords: Neural-Symbolic NLP, LLMs, Explainability

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